HAOS-IIP Scale-Bridge Legitimacy Audit: A Phase 66-Compatible Companion to the Continuum Physics Scaling Roadmap

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Abstract

This companion note freezes the scale-bridge legitimacy audit for HAOS-IIP as a Phase 66-compatible hostile-audit artifact. It is not a new phase, not Phase 67, not a new mechanism, and not a physical correspondence claim. Its purpose is to strengthen the existing Continuum Physics Scaling Roadmap by formalizing how HAOS-IIP evaluates refinement-ordered scaling diagnostics toward continuum feasibility. The central rule is simple: refinement evidence is not continuum proof. The audit defines ledger requirements for scale-facing claims, matched-control scaling comparisons, proto-geometry surrogate language rules, limited metric-like tests, coarse-graining recovery checks, status-box requirements, corrected public language, and explicit failure gates. It also records a v1.2 initial diagnostic extract from frozen ledgers: a Phase XVIII metric-surrogate refinement/control-separation table and a Phase X projection-recovery bookkeeping table. These data do not prove a continuum limit. They only make the scale-bridge evidence easier to inspect, criticize, and bound.

1 Status

This paper is a numbered PDF companion to:

- `docs/notes/foundations/HAOS_IIP_Scale_Bridge_Legitimacy_Audit_v1.md`
- `docs/notes/foundations/HAOS_IIP_Continuum_Physics_Scaling_Roadmap_v1.md`
- `docs/notes/foundations/HAOS_IIP_Phase_66_Translation_and_Hostile_Audit_Layer_v1.md`

It is:

- a Continuum Physics Scaling Roadmap companion artifact;
- a Phase 66-compatible hostile-audit document;
- a scale-bridge legitimacy checklist;
- a public-language correction layer for continuum-facing claims.

It is not:

- a new phase sequence;

- Phase 67;
- a new operator mechanism;
- a proof of a continuum limit;
- a derivation of spacetime, known physics, physical geometry, or physical correspondence.

2 Core Rule

Refinement evidence is not continuum proof.

The scale bridge becomes more legitimate only when it survives stricter bookkeeping, matched controls, refinement, and compression. The audit does not make the bridge more dramatic. It makes it harder to overstate.

3 Current Scale-Bridge Problem

HAOS-IIP has strong internal frozen-regime diagnostics, scalar-carrier geometry closure on one tested kernel-graph family, bounded proto-continuum evidence, and a reproducible late-phase propagation, ordering, causal-closure, and distance-surrogate chain. That is meaningful. It is not yet a proven continuum limit.

The current weakness is the scale bridge. The program can show refinement-ordered behavior and metric-like surrogate stabilization inside tested operator regimes, but it has not shown that these structures converge uniquely, universally, or physically toward a continuum geometry or continuum field theory.

The audit therefore asks:

> Is HAOS-IIP detecting structured refinement behavior, or generic graph behavior with disciplined language?

The honest current status is:

- internal diagnostics are strong inside declared frozen regimes;
- CP4 has a bounded shared-family scalar kernel-graph geometry closure;
- proto-geometric surrogate behavior exists at the tested level;
- CP5 universality is still open;
- continuum ontology is not established.

4 Relation to the CP Ladder

This audit attaches to the existing CP1–CP6 ladder. It does not introduce a separate sequence.

CP gate	Roadmap role	Scale-bridge audit question

CP1	Scalar continuum contract	Does the scalar operator converge beyond one clean substrate family and one normalization?
CP2	Vector continuum contract	Is apparent vector behavior a true low coexact branch or inherited/topological structure?
CP3	Effective-equation contract	Do coarse laws survive refinement without retuning and outperform controls?
CP4	Geometry closure contract	Does one metric-like surrogate organize the same tested branch family without importing geometry by hand?
CP5	Universality contract	Does the effective statement survive nontrivial changes of construction?
CP6	Bounded continuum-physics statement	Can the result be stated as bounded continuum feasibility without ontology creep?

CP1, CP4, and CP5 carry the heaviest scale-bridge load. CP1 controls the scalar continuum foothold. CP4 controls whether geometry-like language is earned inside one shared operator/substrate family. CP5 controls whether the bridge is a one-family artifact or a robust continuum-feasibility claim. If CP5 fails, CP6 must shrink.

5 Refinement Scaling Ledger

Every continuum-facing scale claim should create or reference a public refinement scaling ledger. A clean-looking trend without its failed controls is not scale-bridge evidence.

Field	Meaning
`claim_id`	Stable identifier for the scale-bridge claim
`operator_family`	`L0`, `L1`, `D_H`, block cochain Laplacian, scalar carrier, or declared operator
`substrate_graph_family`	Cubic grid, perturbed grid, random geometric graph, kernel graph, periodic box, or declared family
`refinement_level`	`n_side`, `h`, sample count, or hierarchy index
`kernel_width_scaling_rule`	Fixed width, $\epsilon_k \sim h$, $\epsilon_k \sim h^2$, adaptive width, or other rule
`normalization`	Discrete second-moment normalization, trace normalization, mass normalization, or declared alternative
`measured_observable`	Eigenvalue, trace ratio, Green response, shell slope, metric surrogate, current closure, etc.
`matched_control`	Explicit control family used at the same level
`error_trend`	Refinement-ordered error, residual, drift, or fit gap
`dimensionless_ratio`	Unitless ratio used to compare across refinement
`recoverability_index`	Retention or reconstruction score under perturbation
`branch_identity_persistence`	Overlap, principal angle, label stability, or spectral address persistence

metric_surrogate_stability	Metric-like stability score, bounded violation rate, or ordering stability
status	PASS, OPEN, or FAIL
claim_boundary	What the row explicitly does not claim

6 Matched-Control Scaling Comparison

Every scale-bridge claim must compare the branch result against controls chosen to be close enough to be dangerous:

- random graph control;
- shuffled-weight control;
- matched-degree control;
- non-admissible operator control;
- perturbed-kernel control;
- seed-randomized ensemble.

Control comparison should happen at the same level as the claim. If the claim concerns eigenvalue scaling, compare eigenvalue scaling. If it concerns metric-like behavior, compare metric-like behavior. If it concerns coarse-law closure, compare equation-fit residuals and coefficient drift.

7 Proto-Geometry Surrogate Language

Allowed standard-language terms:

- metric surrogate;
- distance proxy;
- curvature proxy;
- spectral separation;
- subspace overlap decay;
- Green-response distance;
- heat-kernel behavior;
- shell-arrival slope;
- operator-response distance;
- refinement-ordered scaling diagnostic.

Forbidden stronger terms unless later proven by a stronger contract:

- emergent spacetime;

- derived physical geometry;
- physical metric;
- real curvature;
- continuum manifold;
- field-theory derivation.

Correct public wording:

> metric-like surrogate behavior stabilizes under refinement inside the tested operator regime

Incorrect public wording:

> proto-geometry emerges

unless the document has already specified the surrogate, controls, gates, and non-claims.

8 Limited Metric-Like Tests

For each metric surrogate or distance proxy, the document must test or explicitly defer:

- non-negativity;
- identity behavior within numerical tolerance;
- symmetry, when applicable;
- triangle-inequality behavior or bounded violation;
- refinement stability;
- perturbation stability;
- matched-control separation.

Passing these tests means metric-like surrogate behavior stabilizes inside the tested operator regime. It does not mean HAOS-IIP has derived spacetime or a physical metric.

9 Coarse-Graining Recovery

Refinement alone is not enough. The structure must also survive compression. For refinement levels $h_i < h_j$, with projection or coarse-grain maps

$$\Pi_{h_i \to h_j},$$

the audit requires testing whether refined structure projects back to coarser levels while preserving:

- branch identity;
- recoverability ranking;

- spectral address persistence;
- metric-surrogate ordering;
- trace-ratio stability.

Core question:

> Does the structure survive both refinement and compression?

Failure under compression means the apparent refinement trend may be a high-resolution artifact.

10 Scale-Bridge Status Box

Every future continuum-facing document should include this status box:

> **Scale-Bridge Status Box**
>
> What is refined: ...
>
> What stabilizes: ...
>
> What fails: ...
>
> What matched control was used: ...
>
> What continuum-feasibility evidence exists: ...
>
> What is not claimed: ...

The final field is mandatory. This does not claim a proven continuum limit, derivation of space-time, derivation of known physics, physical correspondence, uniqueness of the limit, or replacement of established models.

11 Corrected Language Rules

Replace	Use instead
continuum bridge	refinement-ordered scaling diagnostics toward continuum feasibility
proto-geometry emerges	metric-like surrogate behavior stabilizes under refinement inside the tested operator regime
physical observable	bridge observable candidate
derived metric field	operator-derived metric surrogate

Short phrases are allowed in titles and file names, but the first public explanation must use the corrected language.

12 Failure Gates

A scale-bridge claim fails if:

- measured structure does not stabilize under refinement;
- stabilization is no better than matched controls;
- metric-like behavior depends on hand-picked parameters;
- branch identity changes unpredictably across refinement;
- coarse-graining destroys the claimed structure;
- error trends do not improve with refinement;
- null models reproduce the same effect;
- interpretation requires importing geometry by hand.

When a failure gate triggers, the claim must shrink to the last surviving lower gate. A failed CP5 universality claim may still leave a valid CP4 one-family geometry closure. A failed metric surrogate may still leave valid operator convergence.

13 v1.1 Focused Lift-Cycle Templates

Status:

- focused lift cycle active;
- initial convergence diagnostic extract recorded in Section 14;
- next target: same-surrogate coarse-graining recovery before any stronger scale-bridge statement.

13.1 Minimal convergence table template

Level	Operator	Observable	HAOS value	Control value	Recovery	Status
TBD	TBD	spectral gap / trace ratio	TBD	TBD	TBD	OPEN

Minimum acceptance rule: at least three deterministic refinement levels, one matched control with the same observable and normalization, one coarse-graining recovery score, explicit PASS/OPEN/FAIL, and explicit claim boundary in every row.

13.2 Matched-control failure example template

Control family	What was matched	What failed	Interpretation
TBD	degree / weights / seed / kernel / size	scaling trend / branch identity / recovery	control does not reproduce the claimed refinement-ordered structure

The control failure must state what was held fixed, what diverged, and whether the difference survives normalization.

13.3 Scale-Bridge Status and Limits paragraph template

This artifact tests refinement-ordered scaling diagnostics toward continuum feasibility inside the declared HAOS-IIP operator regime. What is refined is [operator/substrate/level]. What stabilizes is [observable/signature/surrogate]. What fails or remains open is [failure/open boundary]. The matched control is [control family]. The continuum-feasibility evidence is [bounded evidence]. This does not claim a proven continuum limit, derivation of spacetime, derivation of known physics, physical correspondence, uniqueness of the limit, or replacement of established models.

13.4 Bounded predictive bridge result template

Bridge line	Predeclared target	Threshold	Data boundary	Result	Status
ferron / magnon / biology	spectral feature / recovery threshold / target window	TBD	raw-data status	TBD	OPEN

Minimum acceptance rule: target declared before full audit, raw-data boundary stated, failures recorded next to successes, and no language stronger than bridge observable candidate.

14 v1.2 Initial Convergence Diagnostic Extract

This section records the first populated scale-bridge diagnostic extract. It is derived only from already frozen ledgers. It introduces no new dynamics, no new threshold, and no continuum-proof claim.

14.1 Metric-Surrogate Refinement

The first table uses the Phase XVIII distance-surrogate shell slope, `mean_arrival_vs_depth_slope`, reconstructed on the `bias_onset`, ensemble-7 slice. The matched control is `periodic_diagonal_augmented_contr`

N	Branch	Control	Ratio	B drift	C drift	Status
60	1.004762	0.682051	1.473147	baseline	baseline	PASS baseline; coarse-grain OPEN
72	1.004762	1.266667	0.793233	0.000000	0.584615	PASS stability; coarse-grain OPEN
84	1.009524	1.028571	0.981481	0.004762	0.238095	PASS stability; coarse-grain OPEN

The branch shell-slope band remains compact across $N = 60, 72, 84$, with maximum adjacent drift 0.004762. The matched control reaches maximum adjacent slope drift 0.584615. The branch also keeps causal-depth drift at 0.000000, while the matched control reaches 0.714286. This is bounded evidence for stabilized metric-like surrogate behavior inside the tested operator regime. It is not continuum proof.

14.2 Projection-Recovery Compression

The second table records Phase X low-mode spectral projection under cutoff $\lambda \leq 7.5$. It supports compression bookkeeping, not branch-specific universality.

Level	N	B retained %	C retained %	B trace err.	C trace err.	Status
R1	12	96.528	97.917	0.032208	0.019219	PASS recovery; specificity OPEN
R2	24	96.354	97.049	0.033787	0.027282	PASS recovery; specificity OPEN
R3	36	96.219	96.528	0.035043	0.032131	PASS recovery; specificity OPEN
R4	48	95.790	96.398	0.039037	0.033349	PASS recovery; specificity OPEN
R5	60	95.972	96.306	0.037338	0.034210	PASS recovery; specificity OPEN

Low modes and ratio ordering are preserved on both branch and control across R1–R5. The table therefore supports projection recovery bookkeeping, but it does not by itself establish branch-specific scale-bridge separation.

14.3 Matched-Control Separation

Control	What was matched	Branch result	Control result
periodic diagonal augmented control	disturbance, seeds, ensemble size, arrival/depth observable, refinement levels	max slope drift 0.004762; causal-depth drift 0.000000	max slope drift 0.584615; causal-depth drift 0.714286

The control does not reproduce the branch's stabilized metric-surrogate behavior under the same refinement diagnostic.

14.4 Status and Limits

This v1.2 extract tests refinement-ordered scaling diagnostics toward continuum feasibility inside the declared HAOS-IIP operator regime. What is refined is the frozen branch-local cochain-Laplacian hierarchy on the Phase XVIII $N = 60, 72, 84$ distance-surrogate slice and the Phase X R1–R5 spectral-projection slice. What stabilizes is the branch arrival-vs-depth shell slope, causal-depth identity, and low-mode projection bookkeeping. What remains open is same-surrogate coarse-graining recovery, CP1 scalar convergence beyond the existing bounded contract, and CP5 universality across broader substrate/kernel families. This does not claim a proven continuum limit, derivation of spacetime, derivation of known physics, physical correspondence, uniqueness of the limit, or replacement of established models.

15 Final Boundary

Scale-bridge evidence is useful.

Scale-bridge evidence is not continuum proof.

The correct posture is not pessimism. It is recoverable coherence under stronger interaction with controls, refinement, and compression.